Systematic resampling can increase variance

Disproof of Conjecture 00000008263

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Abstract

Four positively weighted particles give a systematic resampling estimator with variance 1/16, while multinomial resampling has variance 3/64. Thus systematic resampling can be worse, and cannot universally halve multinomial variance, whether halving means equality or an upper bound. The exact inverse-CDF algorithms and finite probability calculations are verified in Lean.

The assertion being tested

The source claims a variance-halving law for systematic versus multinomial resampling. We test the usual conditional resampling variance of the equally weighted empirical estimate of a fixed particle observable, conditional on the input particles and normalized weights. A universal variance-halving law would in particular require

$$\operatorname{Var}(\widehat{\mu}_{\mathrm{sys}}(f)) \le \tfrac{1}{2}\operatorname{Var}(\widehat{\mu}_{\mathrm{mult}}(f))$$

for every such input and observable. We give a strict counterexample even to the weaker assertion that systematic variance never exceeds multinomial variance. This refutes the variance conjunct; it is unnecessary to settle the source's effective-sample-size or other error-constant assertions.

This phenomenon is known: systematic resampling does not enjoy a universal variance improvement over multinomial resampling; see Douc, Cappé and Moulines, *Comparison of Resampling Schemes for Particle Filtering* (2005), arXiv:cs/0507025. The present four-particle calculation is supplied as an explicit, independently checkable instance, not as a new general phenomenon.

Particles, ordering, and the actual algorithms

Keep the given particle ordering and choose

$$(w_0, w_1, w_2, w_3) = (1/8, 3/8, 1/8, 3/8), \qquad (f_0, f_1, f_2, f_3) = (1, 0, 1, 0).$$

All weights are strictly positive and their sum is one. The target mean is $\mu(f) = \sum_i w_i f_i = 1/4$. The inverse CDF sends a position $x \in [0, 1)$ to the unique particle whose half-open cumulative-weight interval contains x:

$$L(x) = \begin{cases} 0, & 0 \le x < 1/8, \\ 1, & 1/8 \le x < 1/2, \\ 2, & 1/2 \le x < 5/8, \\ 3, & 5/8 \le x < 1. \end{cases}$$

Systematic resampling chooses one uniform offset $U \in [0, 1/4)$ and selects the labels $L(U + k/4)$, $k = 0, 1, 2, 3$. Its estimator is

$$\widehat{\mu}_{\mathrm{sys}}(f) = \tfrac{1}{4}\sum_{k=0}^{3} f_{L(U+k/4)}.$$

Multinomial resampling chooses four independent labels J_k, each with $\mathbb{P}(J_k = i) = w_i$, and uses $\widehat{\mu}_{\mathrm{mult}}(f) = \frac{1}{4}\sum_{k=0}^{3} f_{J_k}$. The particles may be placed at four distinct state-space points with the displayed observable values, so no coincidence or particle identification is needed.

Exact systematic variance

For every offset, including the interval boundaries under the stated half-open convention, the selected label vector is

$$(L(U), L(U+1/4), L(U+1/2), L(U+3/4)) = \begin{cases} (0,1,2,3), & 0 \le U < 1/8, \\ (1,1,3,3), & 1/8 \le U < 1/4. \end{cases}$$

Each phase interval has probability $1/2$, since it has length $1/8$ inside an interval of length $1/4$. Hence

$$\widehat{\mu}_{\mathrm{sys}}(f) = \begin{cases} 1/2, & \text{with probability } 1/2, \\ 0, & \text{with probability } 1/2. \end{cases}$$

It is unbiased and has variance

$$\tfrac{1}{2}(1/2 - 1/4)^2 + \tfrac{1}{2}(0 - 1/4)^2 = \frac{1}{16}.$$

Exact multinomial variance and contradiction

Each f_{J_k} is a Bernoulli variable with success probability $w_0 + w_2 = 1/4$. Independence gives

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{mult}}(f)) = \frac{1}{4^2}\sum_{k=0}^{3}\frac{1}{4}\left(1 - \frac{1}{4}\right) = \frac{3}{64}.$$

Both estimators therefore have the same correct expectation $1/4$, but

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{sys}}(f)) = \frac{4}{64} > \frac{3}{64} = \mathrm{Var}(\widehat{\mu}_{\mathrm{mult}}(f)).$$

In particular systematic variance is neither equal to nor bounded by one half of multinomial variance. The same strict comparison holds for the unnormalized sum estimator, since both variances are multiplied by 16. An extra assumption imposing a special particle ordering or an additional random permutation would describe a different claim. No such assumption appears in the conjecture; systematic resampling in the stated ordering is the standard algorithm.

Formal verification and scope

Lean defines the actual four-particle inverse-CDF algorithm from its cumulative weights and uses the actual systematic positions $U + k/4$. For an arbitrary ordered field, it proves the selected-label formula above for *every* offset in $[0, 1/4)$. Thus it applies in particular to real offsets, and does not merely evaluate two chosen offsets.

A finite probability law is defined by exact rational masses, with proofs of nonnegativity and total mass one. Expectation is the mass-weighted finite sum and variance is the expectation of the squared deviation from the expectation. The systematic law is uniform on the two phase strata; the estimator is still generated by inverse CDF at a representative of each stratum. The selected-label theorem justifies this representation for every observable. The multinomial law is the genuine product law on all $4^4 = 256$ ordered label tuples, with mass equal to the product of their four input weights.

Lean computes the expectations and centered variances from these algorithms and laws, proves the strict worse-than-multinomial comparison, and negates the necessary universal halving assertion over all rational observables on this fixed legal weighted population. The final theorem is

`Conjecture8263.conjecture8263_false`.

The counterexample observable is rational, so restriction to rational observables cannot hide the failure of a claim over real observables.

The elementary measure identification of a continuous uniform offset with the two equal-length phase strata is supplied on paper. Lean formalizes all offsets' algorithmic outputs and the resulting finite probability calculations, not a general measure-theoretic implementation of uniform real random variables or particle-filter evolution. The finite sums are computations of the actual algorithmic random variables, not an imposed table of variance values.

Source and reproduction

The unmodified bilingual source is supplied as `SOURCE.md`, with frozen-source provenance and SHA-256 in the verification folder. The coordinating workflow found no prior submission for this conjecture. Lean is pinned to 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

Run in the supplied Lean folder:

`lake env lean -DwarningAsError=true Main.lean`.

This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.